

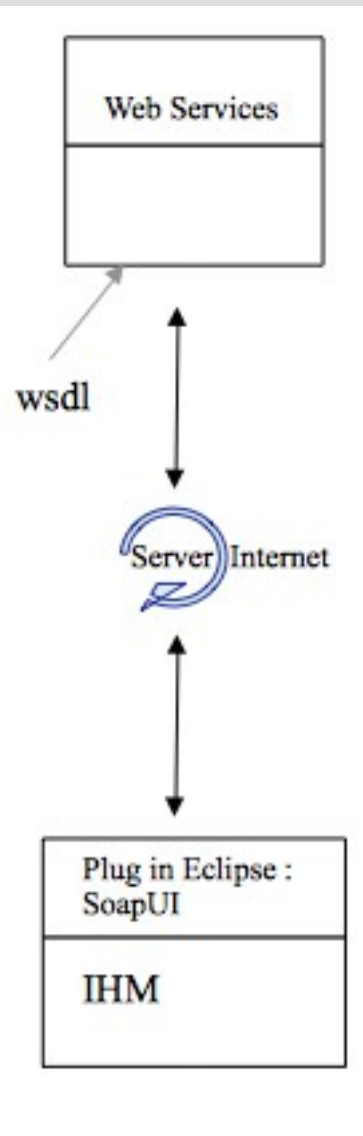
28 Janvier 2014 – Présentation Projet / Nfe204
Le Cnam Hodé Patrice

**WebServices et Base MongoDB
avec les outils Eclipse de
développement :
Plugins SoapUI et MonjaDB**

Sommaire

WebServices et Base MongoDB
avec les outils Eclipse de
développement :
Plugins SoapUI et MonjaDB

Injection de données avec SoapUI



The screenshot shows the SoapUI Eclipse IDE interface. The top toolbar includes icons for Java EE, Java, MonjaDB, Web, and SoapUI. The 'soapUI Navigato' pane on the left shows a project named 'test_soap' with a tree structure containing 'BasicServiceBinding', 'GiveMeBoolean', 'GiveMeDate', 'GiveMeFloat', 'GiveMeInteger', 'GiveMeString', 'RoundFloat', 'StringsAreEqual', 'WhatIsNextDay', and 'HelloSoapBinding'. The 'Request 1' is selected under 'WhatIsNextDay'. The main editor displays the XML request and response for 'Request 1'.

Request XML:

```
<?xml version='1.0' encoding='UTF-8'?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <js:WhatIsNextDay>
      <givenDay>2013-12-20</givenDay>
    </js:WhatIsNextDay>
  </soapenv:Body>
</soapenv:Envelope>
```

Response XML:

```
<?xml version='1.0' encoding='UTF-8'?>
<SOAP-ENV:Envelope xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <SOAP-ENV:Body>
    <SOAP-ENV:WhatIsNextDayResponse>
      <NextDay>2013-12-21T12:00:00</NextDay>
    </SOAP-ENV:WhatIsNextDayResponse>
  </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```

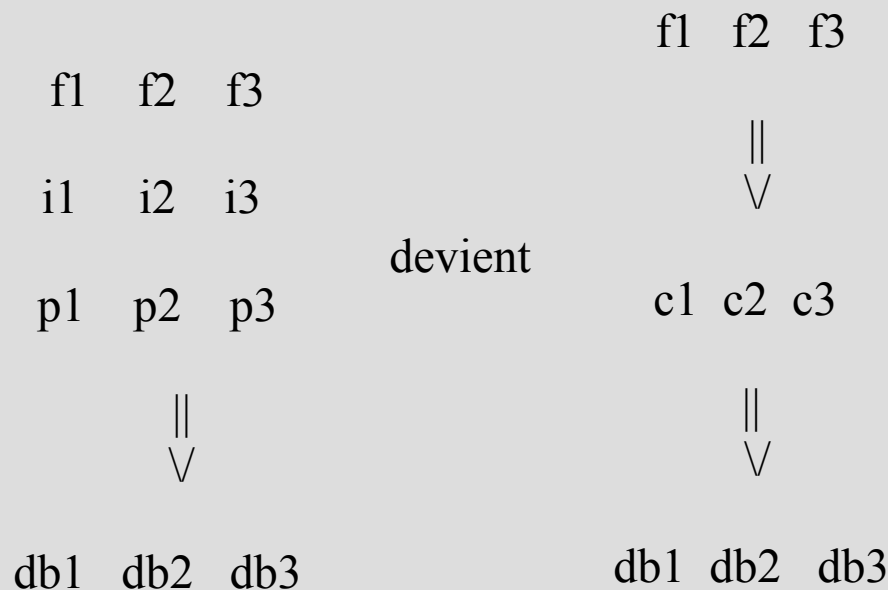
The response time is 279ms (280 bytes). The bottom pane shows the 'soapUI Logs' and 'Console' tabs. The logs show the following messages:

- Mon Jan 20 18:57:21 CET 2014:INFO:Loaded project from [file:/Users/veroniquehode/Documents/Projet_Patrice/test-soap-soapui/]
- Mon Jan 20 18:57:41 CET 2014:DEBUG:runInUiThreadAsync: asyncExec com.eviware.soapui.eclipse.EclipseSoapUIDesktop\$Mode...

Intérêts des WebServices

Chaque fonctionnalité accède à une base de données via des process/interfaces spécifiques

→ Des services métiers réutilisables sont développés et donnent accès à des process et à des bases de données. et cela quelque soit la technologie utilisée par la fonctionnalité et par l'interface : 'principe du Web'

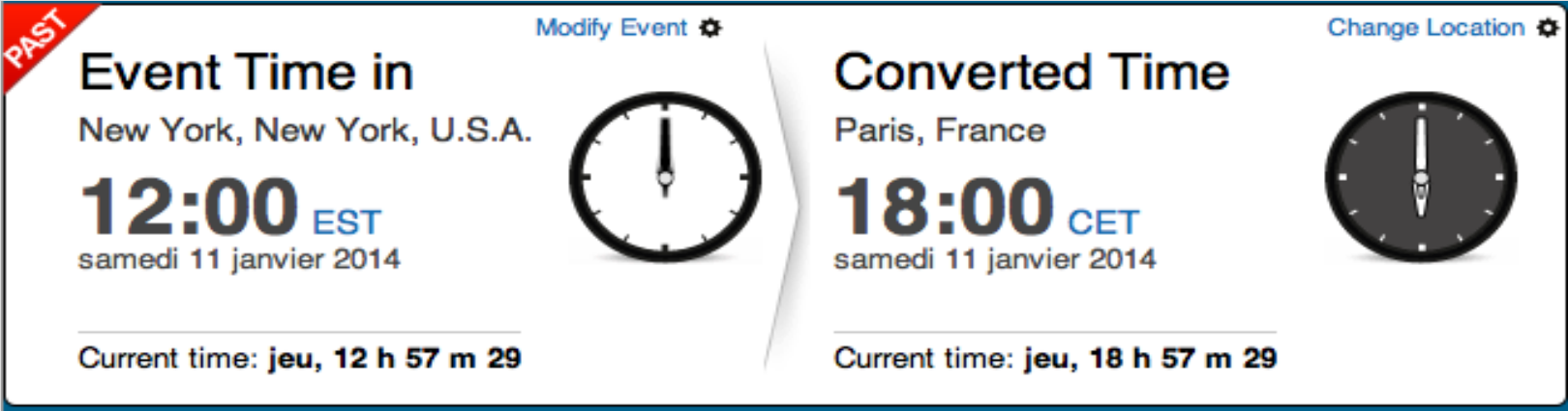


→ simplification
→ standardisation
→ services orientées
architecture (SOA)

Parenthèse : Le WS c'est REST ou SOAP.

REST : contenu dans l'URL

[http://www.timeanddate.com/worldclock/fixedtime.html?
&day=11&month=1&year=2014&hour=12&min=00&sec=0&p1=179](http://www.timeanddate.com/worldclock/fixedtime.html?&day=11&month=1&year=2014&hour=12&min=00&sec=0&p1=179)
→ renvoi une page web avec deux horloges : 12h à nyc =179 et f(lieu)



The screenshot displays a web interface for comparing event times between two locations. On the left, under the heading 'Event Time in', it shows 'New York, New York, U.S.A.' with a large digital clock displaying '12:00 EST' and the date 'samedi 11 janvier 2014'. A small analog clock icon is positioned to the right of the digital time. Below this, it states 'Current time: jeu, 12 h 57 m 29'. On the right, under the heading 'Converted Time', it shows 'Paris, France' with a large digital clock displaying '18:00 CET' and the date 'samedi 11 janvier 2014'. A small analog clock icon is positioned to the right of the digital time. Below this, it states 'Current time: jeu, 18 h 57 m 29'. At the top of the interface, there are links for 'Modify Event' and 'Change Location', each accompanied by a gear icon. A red banner in the top left corner of the interface reads 'PAST'.

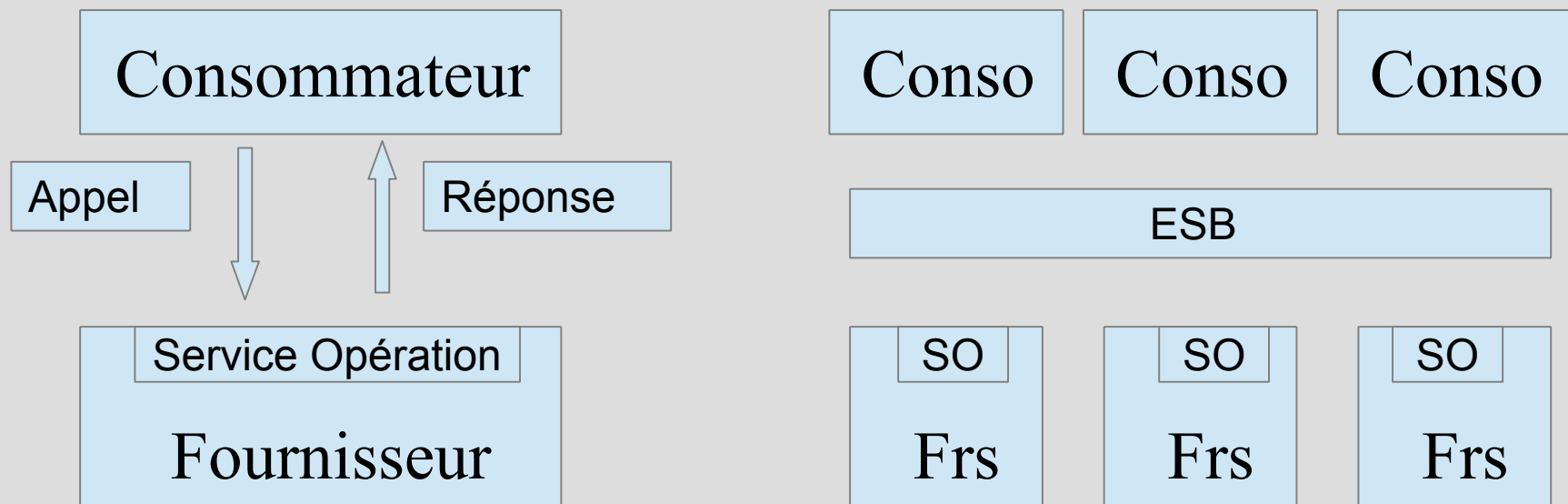
Autre exemple de service REST

[http://www.websvcx.net/CurrencyConvertor.asmx/ConversionRate
?FromCurrency=EUR&ToCurrency=USD](http://www.websvcx.net/CurrencyConvertor.asmx/ConversionRate?FromCurrency=EUR&ToCurrency=USD)

→ renvoi un coefficient 1.3611

Principe des WebServices SOAP

Un système appelle un service dans le but de réaliser une opération



- Bus de services (ESB) (authentification, conformité, routage des requêtes, supervision)
- L'implémentation des données du Fournisseur n'est pas connue du consommateur

Conception des WebServices et des WSDL

Regroupement par données et Regroupement par liens fonctionnels

Fiche de données

....
....
..

Fonctionnalité / IHM

....
....
..

WSDL = Web Service Description Language
Un contrat d'interface écrit en XML
entre le consommateur et le fournisseur

Partie Abstraite

Partie Concrète

Définitions

Types

Messages

...

Port type

Opération's'

...

Binding

Opération's'

Soap:operation

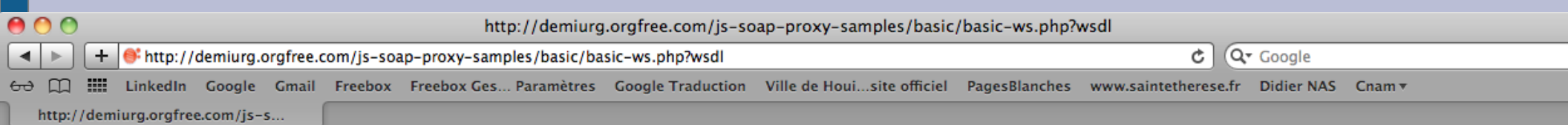
Service

Port

Soap:address

Browser WSDL : adresse + ?WSDL

Partie Abstraite



This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8"?>
<wsdl:definitions xmlns:tns="http://demiurg.orgfree.com/js-soap-proxy-samples/" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/" xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/" xmlns:wsoap="http://schemas.xmlsoap.org/wsdl/"
  name="BasicServiceDefs" targetNamespace="http://demiurg.orgfree.com/js-soap-proxy-samples/">
  <wsdl:types>
    <xsd:schema xmlns="http://www.w3.org/2001/XMLSchema" elementFormDefault="qualified" targetNamespace="http://demiurg.orgfree.com/js-soap-proxy-samples/">
      <!-- ... -->
    </xsd:schema>
  </wsdl:types>
  <wsdl:message name="GiveMeStringSOAPInbound"/>
  <wsdl:message name="GiveMeStringSOAPOutbound"/>
  <wsdl:message name="GiveMeIntegerSOAPInbound"/>
  <wsdl:message name="GiveMeIntegerSOAPOutbound"/>
  <wsdl:message name="GiveMeFloatSOAPInbound"/>
  <wsdl:message name="GiveMeFloatSOAPOutbound"/>
  <wsdl:message name="GiveMeBooleanSOAPInbound"/>
  <wsdl:message name="GiveMeBooleanSOAPOutbound"/>
  <wsdl:message name="GiveMeDateSOAPInbound"/>
  <wsdl:message name="GiveMeDateSOAPOutbound"/>
  <wsdl:message name="StringsAreEqualSOAPInbound"/>
  <wsdl:message name="StringsAreEqualSOAPOutbound"/>
  <wsdl:message name="RoundFloatSOAPInbound"/>
  <wsdl:message name="RoundFloatSOAPOutbound"/>
  <wsdl:message name="WhatIsNextDaySOAPInbound">
    <wsdl:part name="GivenDay" type="xsd:dateTime"/>
  </wsdl:message>
  <wsdl:message name="WhatIsNextDaySOAPOutbound">
    <wsdl:part name="NextDay" type="xsd:dateTime"/>
  </wsdl:message>
  <wsdl:portType name="BasicServicePort">
    <wsdl:operation name="GiveMeString"/>
    <wsdl:operation name="GiveMeInteger"/>
    <wsdl:operation name="GiveMeFloat"/>
    <wsdl:operation name="GiveMeBoolean"/>
    <wsdl:operation name="GiveMeDate"/>
    <wsdl:operation name="StringsAreEqual"/>
    <wsdl:operation name="RoundFloat"/>
    <wsdl:operation name="WhatIsNextDay">
      <wsdl:input message="tns:WhatIsNextDaySOAPInbound"/>
      <wsdl:output message="tns:WhatIsNextDaySOAPOutbound"/>
    </wsdl:operation>
  </wsdl:portType>
</wsdl:definitions>
```

Définitions

Types

Messages

...

Port type

Opération's'

...

Binding

Opération's'

Soap:operation

Service

Port

Soap:address

Partie concrète

```
▼ <wsdl:binding name="BasicServiceBinding" type="tns:BasicServicePort">
  <soap:binding style="rpc" transport="http://schemas.xmlsoap.org/soap/http"/>
  ▶ <wsdl:operation name="GiveMeString">...</wsdl:operation>
  ▶ <wsdl:operation name="GiveMeInteger">...</wsdl:operation>
  ▶ <wsdl:operation name="GiveMeFloat">...</wsdl:operation>
  ▶ <wsdl:operation name="GiveMeBoolean">...</wsdl:operation>
  ▶ <wsdl:operation name="GiveMeDate">...</wsdl:operation>
  ▶ <wsdl:operation name="StringsAreEqual">...</wsdl:operation>
  ▶ <wsdl:operation name="RoundFloat">...</wsdl:operation>
  ▼ <wsdl:operation name="WhatIsNextDay">
    <soap:operation soapAction="http://demiurg.orgfree.com/js-soap-proxy-samples/basic/basic-ws.php#WhatIsNextDay" />
    ▼ <wsdl:input>
      <soap:body use="literal"/>
    </wsdl:input>
    ▼ <wsdl:output>
      <soap:body use="literal"/>
    </wsdl:output>
  </wsdl:operation>
</wsdl:binding>
▼ <wsdl:service name="BasicService">
  ▼ <wsdl:port name="BasicServicePort" binding="BasicServiceBinding">
    <soap:address location="http://demiurg.orgfree.com/js-soap-proxy-samples/basic/basic-ws.php" />
  </wsdl:port>
</wsdl:service>
</wsdl:definitions>
```

Définitions

Types

Messages

...

Port type

Opération's'

...

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Operation's'

Soap:operation

Service

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Soap:address

Sommaire



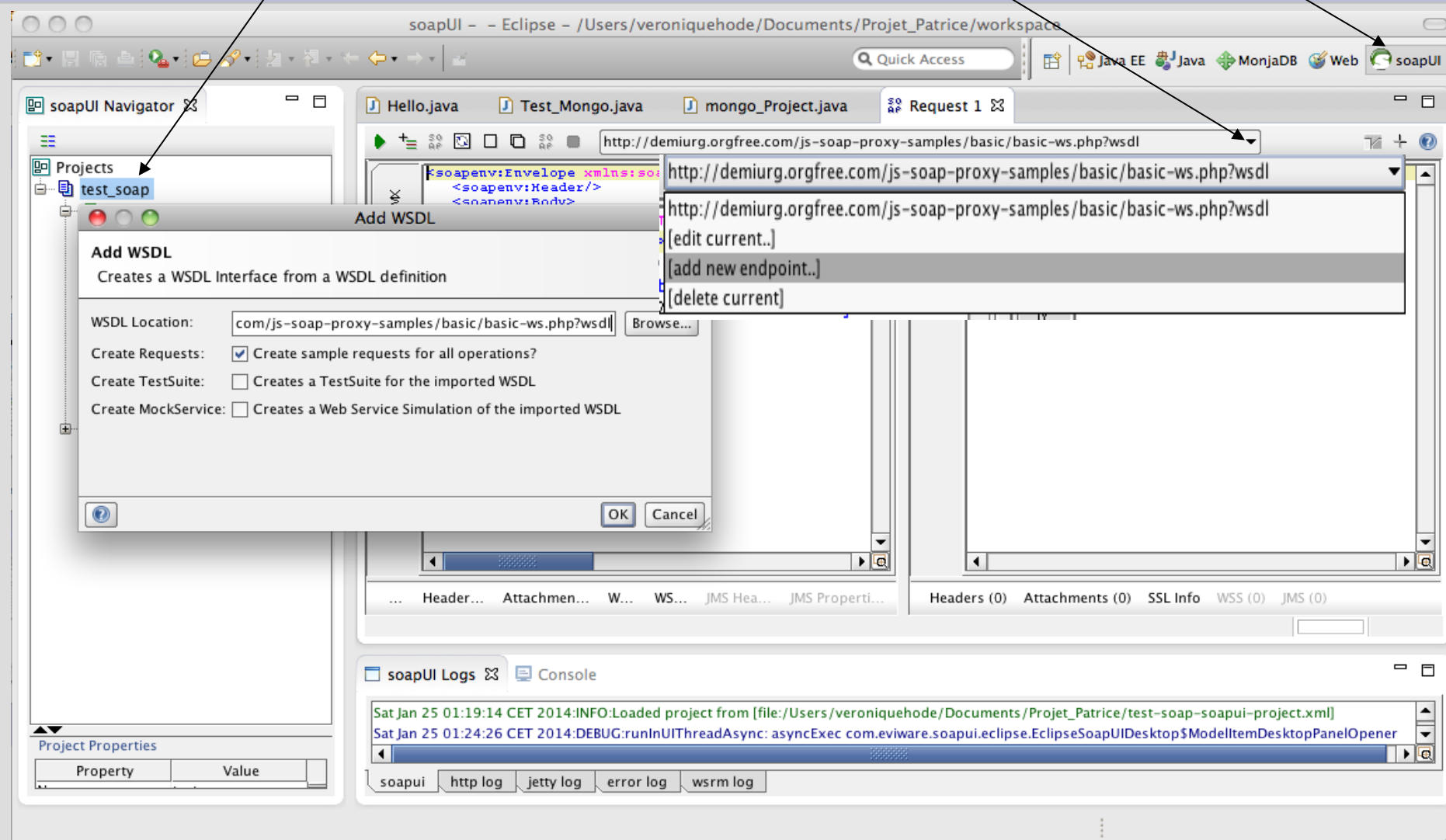
WebServices et Base MongoDB
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Plugins SoapUI et MonjaDB



La perspective SOAPUI

Ajout d'un WSDL et du EndPoint

<http://demiurg.orgfree.com/js-soap-proxy-samples/basic/basic-ws.php>



<http://demiurg.orgfree.com/js-soap-proxy-samples/basic/basic-ws.php>

WSDL Opération WhatIsNextDay 2013-12-20 renvoie 2013-12-21

The screenshot displays the soapUI Eclipse IDE interface. The top toolbar includes icons for file operations and a 'Quick Access' search bar. The main workspace is divided into several panes:

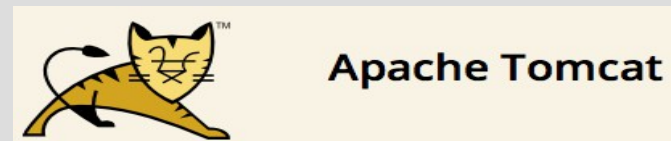
- soapUI Navigator:** Shows a project tree with 'test_soap' containing 'BasicServiceBinding' and 'HelloSoapBinding'. 'Request 1' is selected under 'BasicServiceBinding'.
- Request Editor:** Displays the SOAP request XML for the 'WhatIsNextDay' operation. The XML is as follows:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <js:WhatIsNextDay>
      <GivenDay>2013-12-20</GivenDay>
    </js:WhatIsNextDay>
  </soapenv:Body>
</soapenv:Envelope>
```
- Response Editor:** Displays the SOAP response XML. The XML is as follows:

```
<SOAP-ENV:Envelope xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Body>
    <SOAP-ENV:WhatIsNextDayResponse>
      <NextDay>2013-12-21T12:00:00</NextDay>
    </SOAP-ENV:WhatIsNextDayResponse>
  </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```
- Log Console:** Shows the execution logs, including a message: 'Mon Jan 20 18:57:21 CET 2014:INFO:Loaded project from (file:/Users/veroniquehode/Documents/Projet_Patrice/test-soap-soapui.eclipse.EclipseSoapUIDesktop\$Mode)'. Below the logs are tabs for 'soapui', 'http log', 'jetty log', 'error log', and 'wsrm log'.

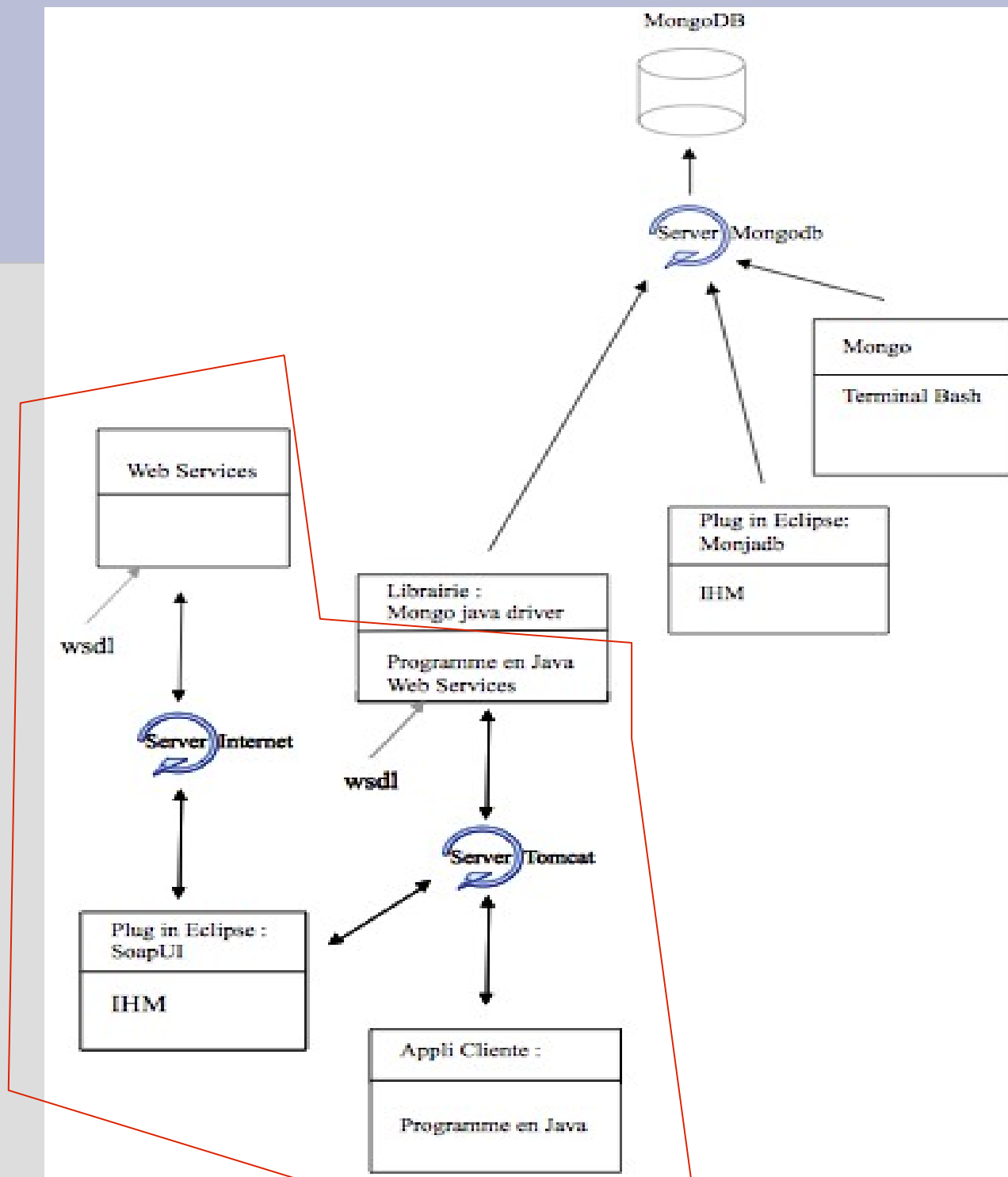
The status bar at the bottom indicates a 'response time: 279ms (280 bytes)' and a duration of '8 : 10'.

Sommaire



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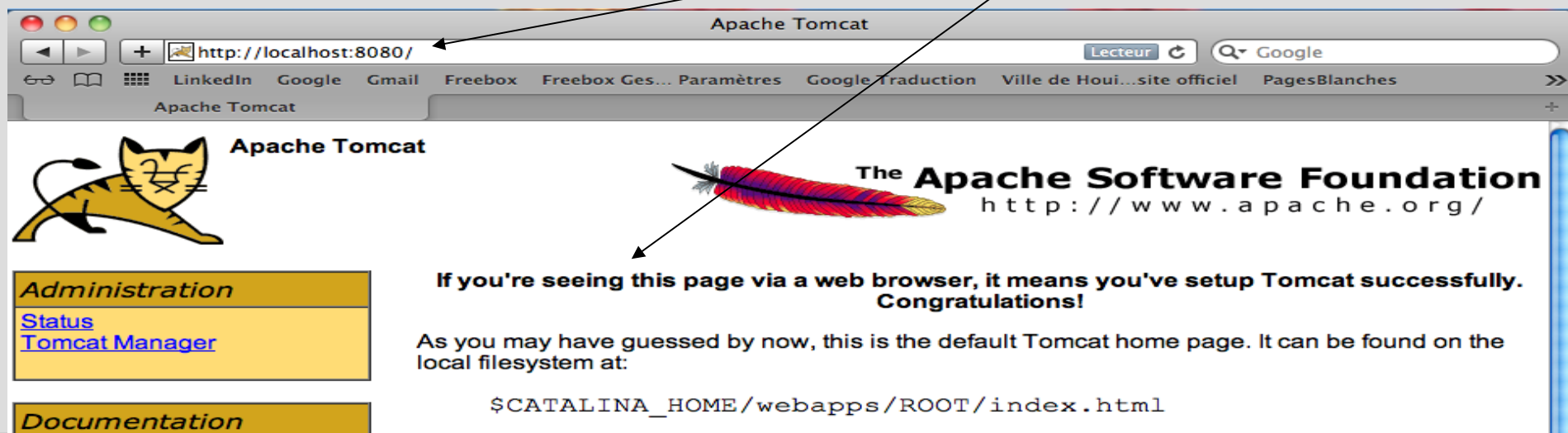
Lancement Serveur Tomcat

Vérification log

et browser local <http://localhost:8080/>

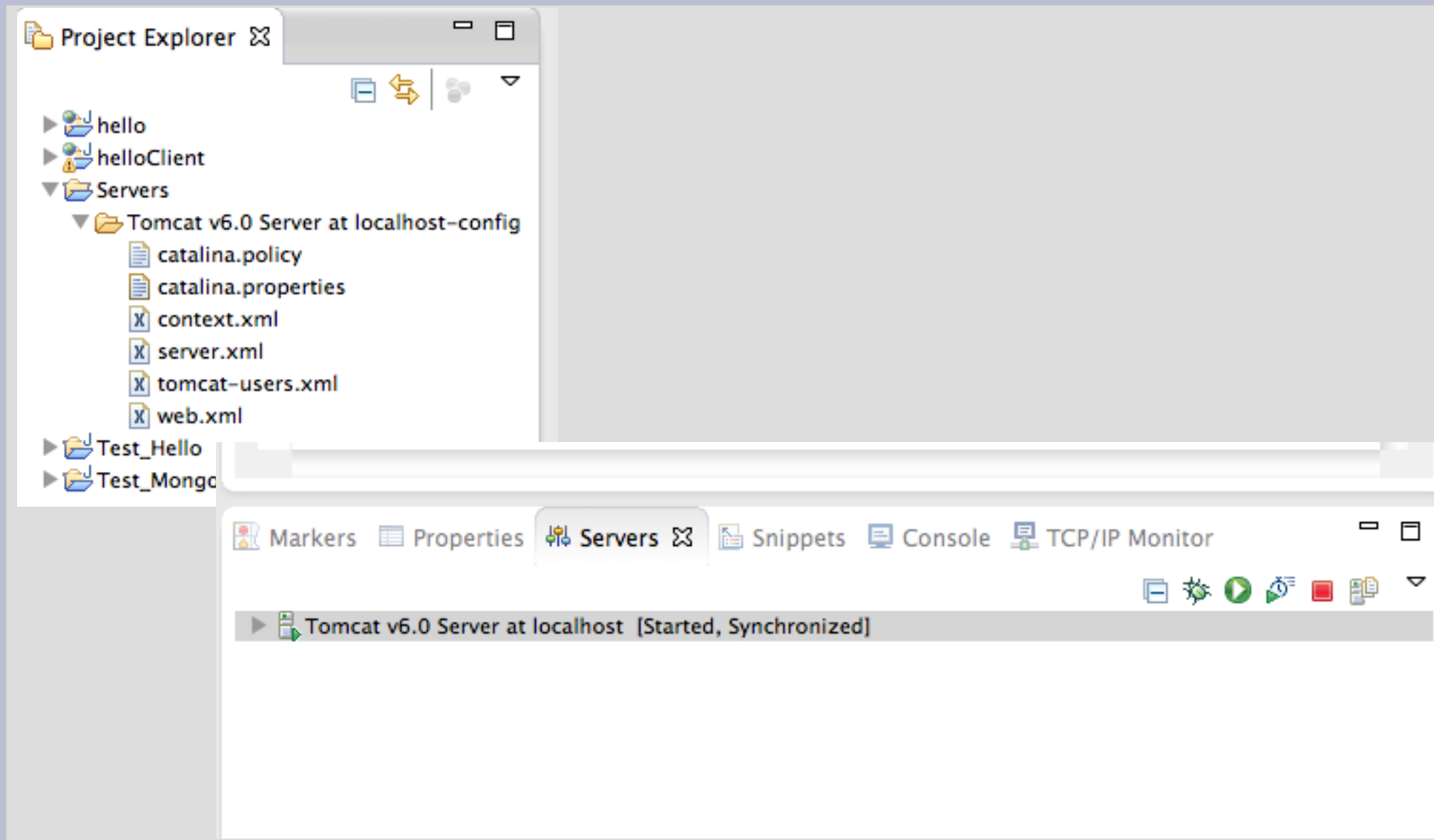
```
Terminal — java — 136x25
Last login: Sun Jan 26 23:48:41 on ttys000
MacBook-Pro-de-veronique-Hode:~ veroniquehode$ cd /Library/Tomcat/Home/bin
MacBook-Pro-de-veronique-Hode:bin veroniquehode$ ./startup.sh
Using CATALINA_BASE:   /Library/Tomcat/Home
Using CATALINA_HOME:   /Library/Tomcat/Home
Using CATALINA_TMPDIR: /Library/Tomcat/Home/temp
Using JRE_HOME:        /System/Library/Java/JavaVirtualMachines/1.6.0.jdk/Contents/Home
Using CLASSPATH:       /Library/Tomcat/Home/bin/bootstrap.jar
MacBook-Pro-de-veronique-Hode:bin veroniquehode$ cat ../logs/catalina.out
INFO: JK: ajp13 listening on /0.0.0.0:8009
27 janv. 2014 18:49:21 org.apache.jk.server.JkMain start
INFO: Jk running ID=0 time=0/25  config=null
27 janv. 2014 18:49:21 org.apache.catalina.startup.Catalina start
INFO: Server startup in 1754 ms
MacBook-Pro-de-veronique-Hode:bin veroniquehode$
```

Lancement Ok



Lancement Serveur Tomcat sur Eclipse

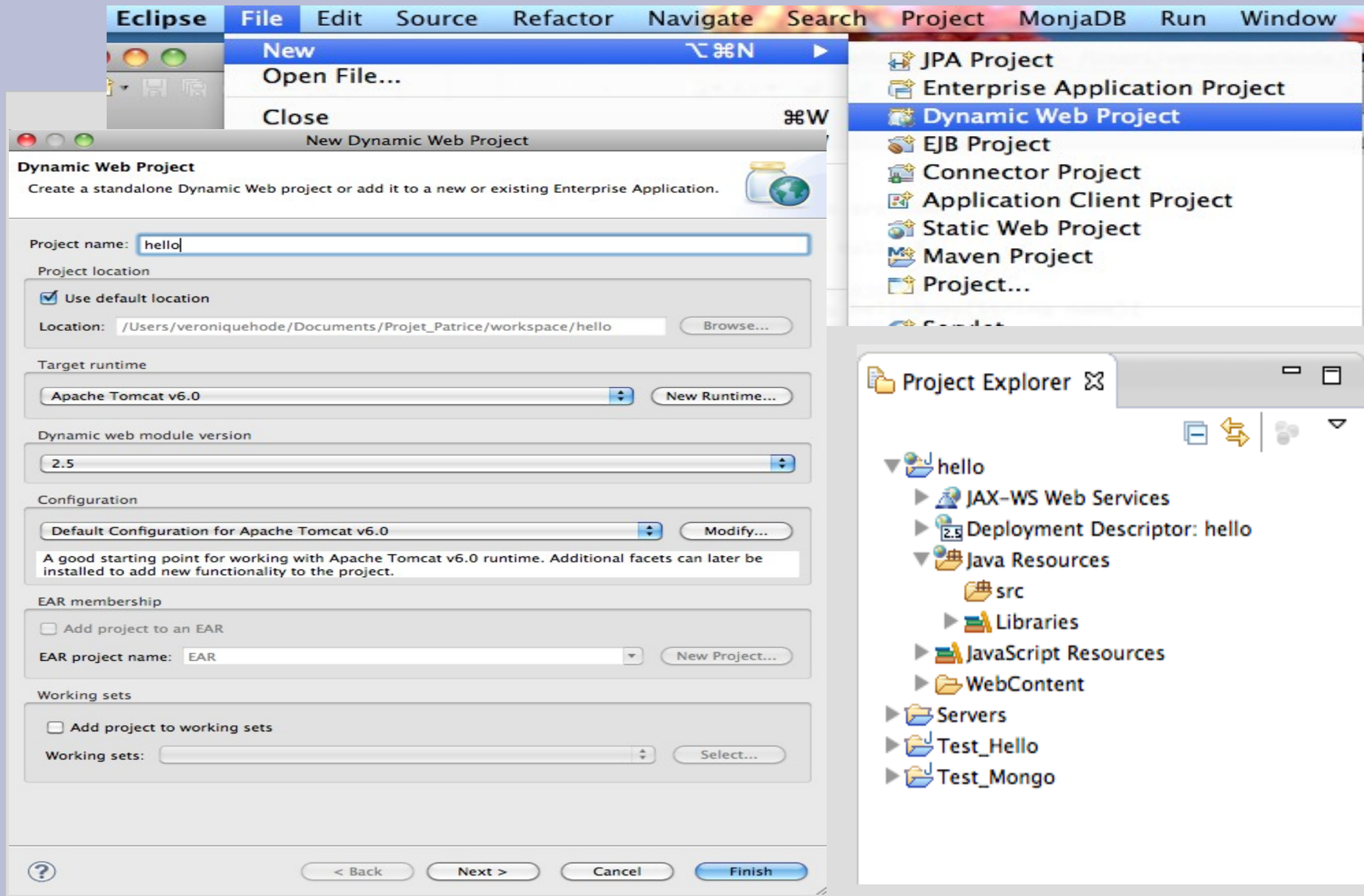
Vérification log et browser local <http://localhost:8080/>



Exemple complet de création d'un Web Service avec Eclipse EE

http://www.eclipse.org/webtools/community/education/web/t320/Implementing_a_Simple_Web_Service.pdf

Création d'un projet Web Dynamique



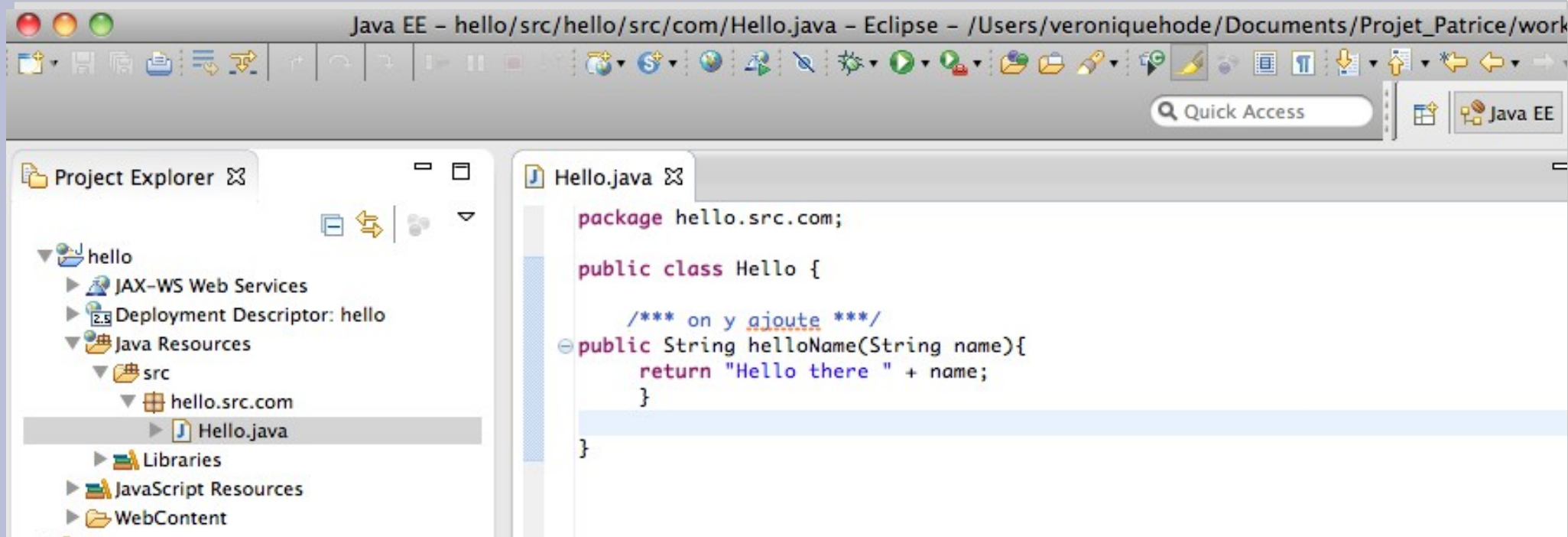
Création d'une classe

The screenshot shows the Eclipse IDE interface. The 'Project Explorer' on the left displays a project named 'hello' with a 'src' folder. A right-click context menu is open over the 'src' folder, with the 'New' option selected. The 'New' submenu is open, showing various options, with 'Class' highlighted. The 'New Java Class' dialog is open on the right, titled 'Java Class'. It contains the following fields and options:

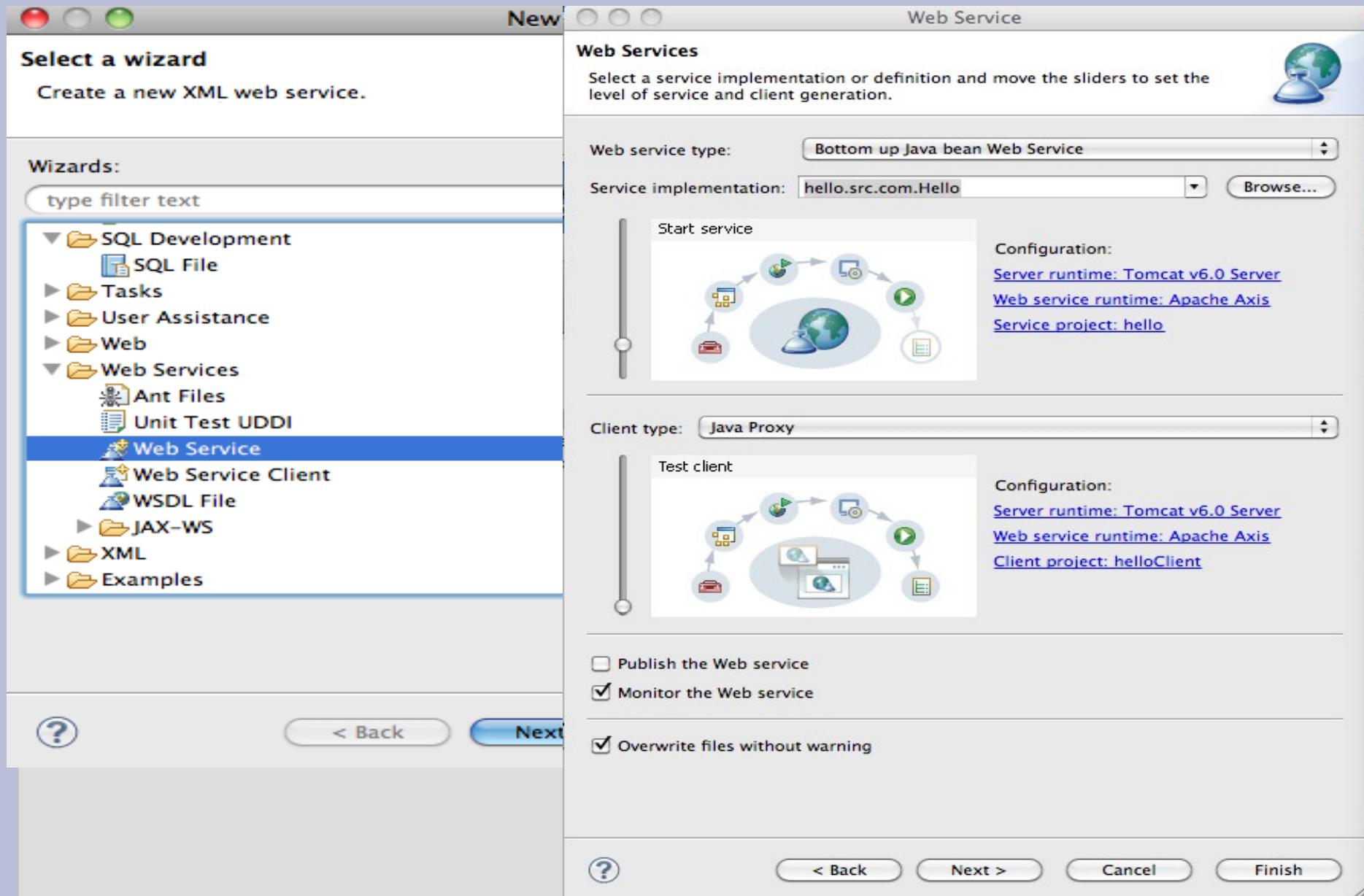
- Source folder:** hello/src (with a 'Browse...' button)
- Package:** hello.src.com (with a 'Browse...' button)
- Enclosing type:** (empty field with a 'Browse...' button)
- Name:** Hello
- Modifiers:** ☒ public, ☐ default, ☐ private, ☐ protected. Below these are checkboxes for ☐ abstract, ☐ final, and ☐ static.
- Superclass:** java.lang.Object (with a 'Browse...' button)
- Interfaces:** (empty list with an 'Add...' button and a 'Remove' button)
- Which method stubs would you like to create?**
 - ☐ public static void main(String[] args)
 - ☐ Constructors from superclass
 - ☒ Inherited abstract methods
- Do you want to add comments?** (Configure templates and default value [here](#))
 - ☐ Generate comments

At the bottom of the dialog are buttons for '?', 'Cancel', and 'Finish'.

Modification de la classe



Création du Web service



Obtention du WSDL et d'une page de test

Java EE – http://localhost:8080/helloClient/sampleHelloProxy/TestClient.jsp?endpoint=http://localhost:12462/hello/services/Hello – Eclipse – /U

Project Explorer

- hello
 - JAX-WS Web Services
 - Deployment Descriptor: hello
 - Java Resources
 - src
 - hello.src.com
 - Hello.java
 - Libraries
 - JavaScript Resources
 - WebContent
 - META-INF
 - WEB-INF
 - wsdl
 - Hello.wsdl
- helloClient
 - JAX-WS Web Services
 - Deployment Descriptor: helloClient
 - Java Resources
 - src
 - hello.src.com
 - Hello.java
 - HelloProxy.java
 - HelloService.java
 - HelloServiceLocator.java
 - HelloSoapBindingStub.java
 - Libraries
 - JavaScript Resources
 - WebContent
 - ECMAScript Built-In Library
 - ECMA 3 Browser Support Library
 - WebContent
 - Servers

Web Services Test Client

http://localhost:8080/helloClient/sampleHelloProxy/TestClient.jsp?endpoint=http://localhost:12462/hello/ser

Methods

- [getEndpoint\(\)](#)
- [setEndpoint\(java.lang.String\)](#)
- [getHello\(\)](#)
- [helloName\(java.lang.String\)](#)

Inputs

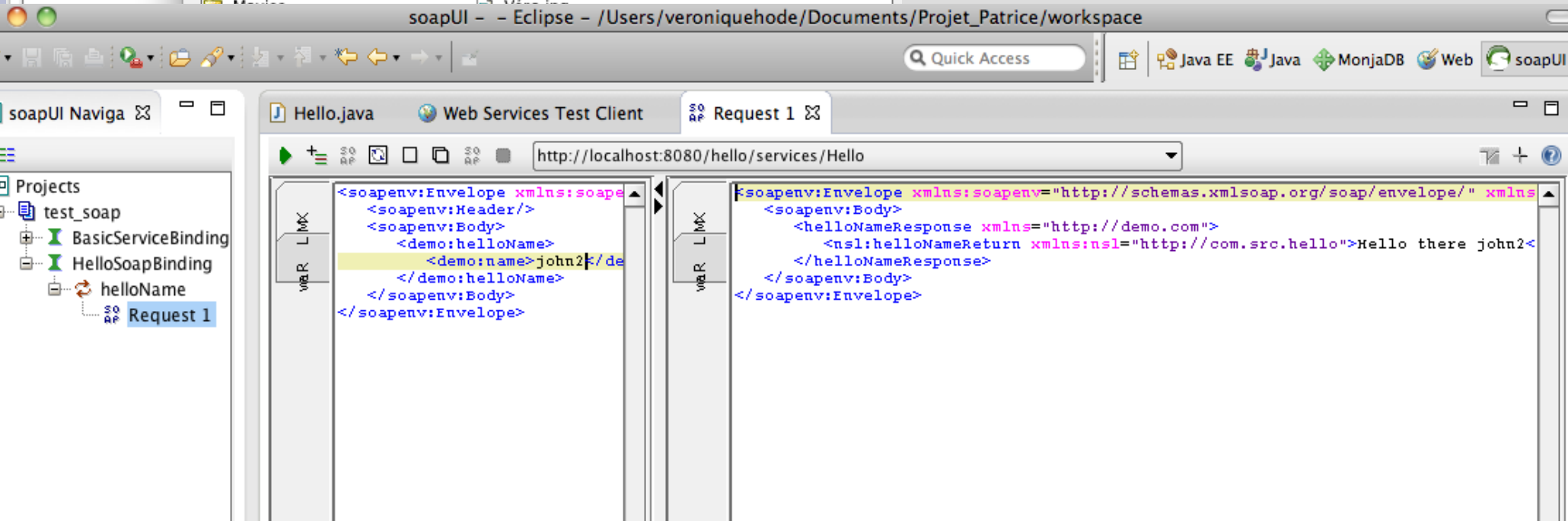
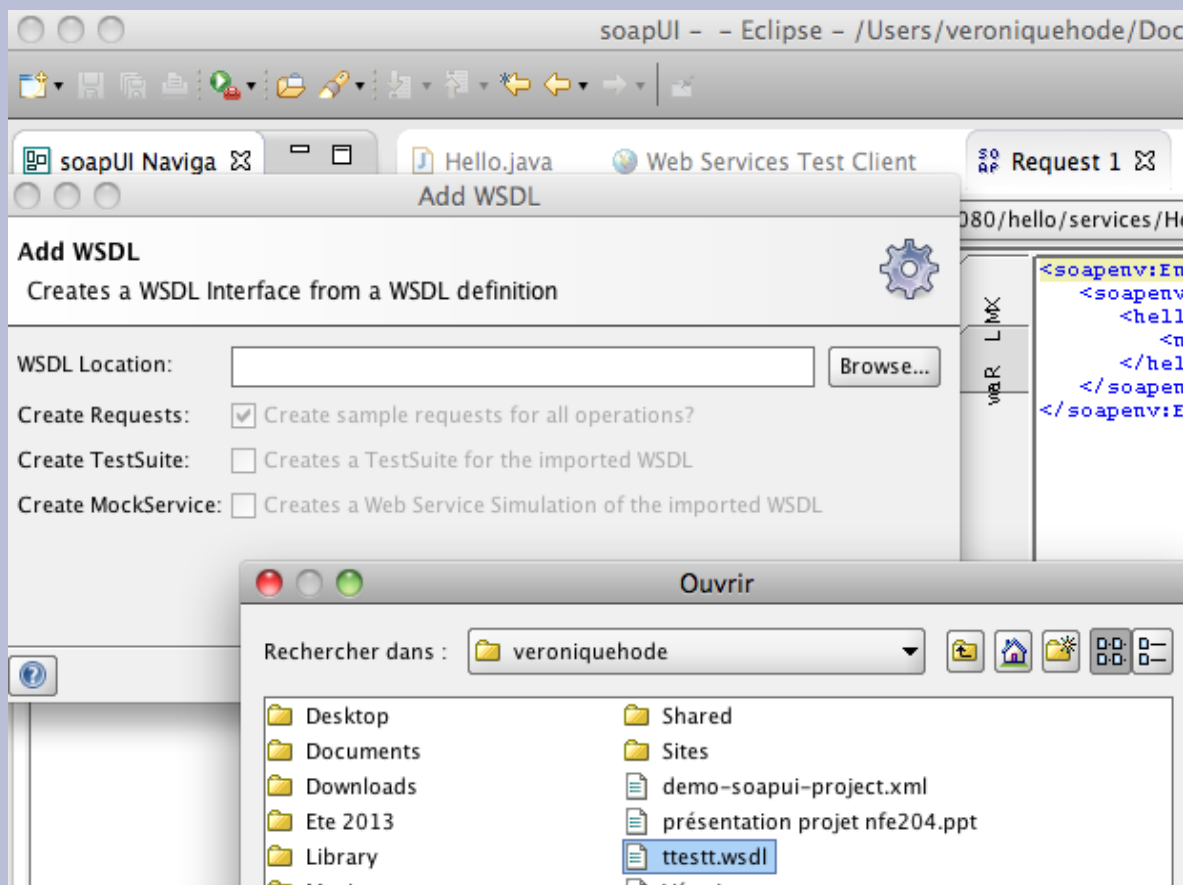
name:

Result

Hello there John

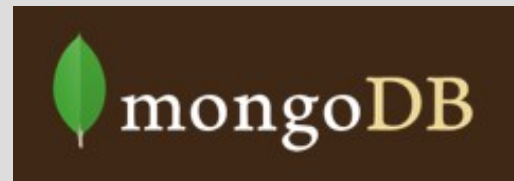
Markers Properties Servers Snippets Console TCP/IP Monitor

Essai avec SoapUI du WSDL en local



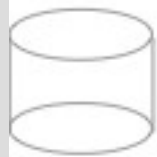
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Lancement du serveur MongoDB

MongoDB



Server Mongod

Mongo

Terminal Bash

Plug in Eclipse:
Monjadb

IHM

Terminal — top — 103x24

Processes: 69 total, 2 running, 67 sleeping, 401 threads
Load Avg: 0.34, 0.23, 0.19 CPU usage: 3.80% user, 4.76% sys, 91.42% idle
SharedLibs: 4920K resident, 4180K data, 0B linkedit.
MemRegions: 13722 total, 984M resident, 26M private, 532M shared.
PhysMem: 573M wired, 1718M active, 1516M inactive, 3808M used, 287M free.
VM: 155G vsize, 1035M framework vsize, 5031618(0) pageins, 810(0) pageouts.
Networks: packets: 3513561/1050M in, 6556115/8450M out. Disks: 274178/12G read, 205301/2993M written.

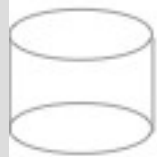
PID	COMMAND	%CPU	TIME	#TH	#WQ	#POR	#MREG	RPRVT	RSHRD	RSIZE	VPRVT	VSIZE	PGRP	PPID
1497	screen captur	0.3	00:00.08	2	1	41	80	456K	13M	2820K	12M	2663M	92	1
1489	mongod	0.6	00:01.24	11	0	47	59	28M+	244K	33M+	94M	2636M	1489	1486
1486	bash	0.0	00:00.00	1	0	17	24	384K	856K	1056K	17M	2378M	1486	1485
1485	login	0.0	00:00.01	1	0	22	53	468K	312K	1576K	19M	2379M	1485	1467
1483	mdworker	0.0	00:00.00	2	1	50	60	1450K	11M	3840K	31M	2413M	1483	1

Terminal — mongod — 104x22

Last login: Fri Jan 24 23:57:39 on ttys001
MacBook-Pro-de-veronique-Hode:~ veroniquehode\$ /Applications/mongodb/bin/mongod ; exit;
/Applications/mongodb/bin/mongod --help for help and startup options
Fri Jan 24 23:57:55.264 kern.sched unavailable
Fri Jan 24 23:57:55.271 [initandlisten] MongoDB starting : pid=1489 port=27017 dbpath=/data/db/ 64-bit host=MacBook-Pro-de-veronique-Hode.local
Fri Jan 24 23:57:55.271 [initandlisten]
Fri Jan 24 23:57:55.271 [initandlisten] ** WARNING: soft rlimits too low. Number of files is 256, should be at least 1000
Fri Jan 24 23:57:55.271 [initandlisten] db version v2.4.8
Fri Jan 24 23:57:55.271 [initandlisten] git version: a350fc38922fbda2cec8d5dd842237b904eafc14
Fri Jan 24 23:57:55.271 [initandlisten] build info: Darwin bs-osx-106-x86-64-2.10gen.cc 10.8.0 Darwin Kernel Version 10.8.0: Tue Jun 7 16:32:41 PDT 2011; root:xnu-1504.15.3~1/RELEASE_X86_64 x86_64 BOOST_LIB_VERSION=1_49
Fri Jan 24 23:57:55.271 [initandlisten] allocator: system
Fri Jan 24 23:57:55.271 [initandlisten] options: {}
Fri Jan 24 23:57:55.272 [initandlisten] journal dir=/data/db/journal
Fri Jan 24 23:57:55.272 [initandlisten] recover : no journal files present, no recovery needed
Fri Jan 24 23:57:55.309 [websvr] admin web console waiting for connections on port 28017
Fri Jan 24 23:57:55.309 [initandlisten] waiting for connections on port 27017

Depuis le terminal Mongo

MongoDB



Server Mongod

Mongo

Terminal Bash

Plug in Eclipse:
Monjadb

IHM

Terminal — mongo — 104x46

```
> // On efface les bases de données
> db.nfe204.remove()
> db.person.remove()
> db.test.remove()
> //on insère
> db.test.insert ({ "nom": "nfe024" })
> //On affiche le contenu de la collection
> db.test.find()
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464b"), "nom" : "nfe024" }
> //On insère d'autre document
> db.test.insert ({ "produit": "Grulband", prix: 230, enStock: true })
> //On peut donner un identifiant explicite :
> db.test.insert ({_id: "1", "produit": "Kramölk", prix: 10, enStock: true})
> //On peut compter
> db.test.count()
3
> //on peut ajouter des enreg
> for(i = 1; i < 21; i++) db.test.save({ username: 'user_' + i, age: i });
> //On affiche le contenu de la collection
> db.test.find()
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464b"), "nom" : "nfe024" }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464c"), "produit" : "Grulband", "prix" : 230, "enStock" : true }
{ "_id" : "1", "produit" : "Kramölk", "prix" : 10, "enStock" : true }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464d"), "username" : "user_1", "age" : 1 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464e"), "username" : "user_2", "age" : 2 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464f"), "username" : "user_3", "age" : 3 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4650"), "username" : "user_4", "age" : 4 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4651"), "username" : "user_5", "age" : 5 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4652"), "username" : "user_6", "age" : 6 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4653"), "username" : "user_7", "age" : 7 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4654"), "username" : "user_8", "age" : 8 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4655"), "username" : "user_9", "age" : 9 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4656"), "username" : "user_10", "age" : 10 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4657"), "username" : "user_11", "age" : 11 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4658"), "username" : "user_12", "age" : 12 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4659"), "username" : "user_13", "age" : 13 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e465a"), "username" : "user_14", "age" : 14 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e465b"), "username" : "user_15", "age" : 15 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e465c"), "username" : "user_16", "age" : 16 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e465d"), "username" : "user_17", "age" : 17 }
Type "it" for more
> it
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e465e"), "username" : "user_18", "age" : 18 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e465f"), "username" : "user_19", "age" : 19 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4660"), "username" : "user_20", "age" : 20 }
>
```

Sommaire



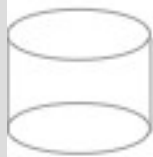
WebServices et Base MongoDB
avec les outils Eclipse de
développement :
Plugins SoapUI et MonjaDB



La perspective MonjaDB

Connection à la base Test de MongoDB

MongoDB



Server Mongod

Mongo

Terminal Bash

Plug in Eclipse:
Monjadb

IHM

Eclipse File Edit Source Refactor Navigate Search Project **MonjaDB** Run Window Help

MonjaDB - Test_Mongo/src/com/demo/Test_Mongo.java

Connect
Disconnect
Preferences

DB Tree DB List Document List Collection List

Connection Configuration

Connection Type

☒ Normal
☐ Via SSH

SSH Configuration

SSH "user@host(:port)"

SSH Private Key File

/Users/veroniquehode

MongoDB

Host : 127.0.0.1

Port : 27017

Database : test

Cancel OK

Actions

Action	Date
mj edit 52b6c915036460274514ecfd	Sun Dec 22 12:...
mj edit 52b6cd5d0364ae27fe431956	Sun Dec 22 12:...
mj edit 52b6cd6003644bcb03b076f7	Sun Dec 22 12:...
dh person find()	Sun Dec 22 12:...

Execute

Saved Actions

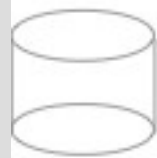
JavaScript on MongoDB Server

Console

MonjaDB Log

Visualisation de la base Mongo

MongoDB



Server Mongod

Mongo

Terminal Bash

Plug in Eclipse:
Monjadb

IHM

Eclipse IDE interface showing the MongoDB plugin integration.

DB Tree: 127.0.0.1:27017

- local
- nfe204
- test
 - person
 - system.indexes
 - test

DB List: db.test.find()

_id	nom	produit	prix	enStock	username	age
52e2f3b...	nfe024					
52e2f3b...		Grulband	230.0	true		
52e2f3b...		Kramölk	10.0	true		
52e2f3b...					user_1	1.0
52e2f3b...					user_2	2.0
52e2f3b...					user_3	3.0
52e2f3b...					user_4	4.0
52e2f3b...					user_5	5.0
52e2f3b...					user_6	6.0
52e2f3b...					user_7	7.0
52e2f3b...					user_8	8.0
52e2f3b...					user_9	9.0
52e2f3b...					user_10	10.0
52e2f3b...					user_11	11.0
52e2f3b...					user_12	12.0
52e2f3b...					user_13	13.0

Document Editor: JSON

```
{  "_id" : ObjectId("52e2f3b8cdbf70bf8e..."),  "produit" : "Grulband",  "prix" : 230,  "enStock" : true}
```

Actions:

Action	Date
mj edit 52e2f3b8cdbf70bf8e464b	Sat Jan 25 00:4...
mj edit 52e2f3b8cdbf70bf8e464c	Sat Jan 25 00:4...
mj edit 52e2f3b8cdbf70bf8e464b	Sat Jan 25 00:4...
mj edit 52e2f3b8cdbf70bf8e464c	Sat Jan 25 00:4...

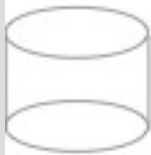
JavaScript on MongoDB Server Console:

MonjaDB Log

```
Sat Jan 25 00:45:27 CET 2014:(MActionManager.java:59)
Sat Jan 25 00:45:27 CET 2014:(MAuthManager.java:28)::
Sat Jan 25 00:45:27 CET 2014:(MAuthManager.java:28)::
Sat Jan 25 00:45:28 CET 2014:(MActionManager.java:59)
Sat Jan 25 00:45:28 CET 2014:(MAuthManager.java:28)::
Sat Jan 25 00:45:28 CET 2014:(MAuthManager.java:28)::
```

Modification de la base Mongo

MongoDB



Server Mongod

Mongo

Terminal Bash

Plug in Eclipse:
Monjadb

IHM

MonjaDB - Test_Mongo/src/com/demo/Test_Mongo.java - Eclipse - /Users/veroniquehode/Documents/Projet_Patrice/workspace

DB Tree

- 127.0.0.1:27017
 - local
 - nfe204
 - test
 - person
 - system.indexes
 - test

Document List

_id	nom	produit	prix	enStock	username	age
52e2f3b...	nfe024					
52e2f3b...		Grulband	240	true		
1		Kramölk	10.0	true		
52e2f3b...					user_1	1.0
52e2f3b...					user_2	2.0
52e2f3b...					user_3	3.0
52e2f3b...					user_4	4.0
52e2f3b...					user_5	5.0
52e2f3b...					user_6	6.0
52e2f3b...					user_7	7.0
52e2f3b...					user_8	8.0
52e2f3b...					user_9	9.0
52e2f3b...					user_10	10.0
52e2f3b...					user_11	11.0
52e2f3b...					user_12	12.0
52e2f3b...					user_13	13.0

Document Editor

```
{
  "_id" : ObjectId("52e2f3b8cdbf70bf8e464b"),
  "produit" : "Grulband",
  "prix" : 230,
  "enStock" : true
}
```

Actions

Action	Date
db.test.find()	Sat Jan 25 00:5...
mj edit 52e2f3b8cdbf70bf8e464b	
mj edit 52e2f3b8cdbf70bf8e464b	
mj edit 52e2f3b8cdbf70bf8e464b	

Saved Actions

JavaScript on MongoDB Server

Console

MonjaDB Log

Terminal — mongo — 104x46

```
>
> // nouvelle visualisation
> db.test.find()
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464b"), "nom" : "nfe024" }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464c"), "produit" : "Grulband", "prix" : 240, "enStock" : true }
{ "_id" : "1", "produit" : "Kramölk", "prix" : 10, "enStock" : true }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464d"), "username" : "user_1", "age" : 1 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464e"), "username" : "user_2", "age" : 2 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e464f"), "username" : "user_3", "age" : 3 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4650"), "username" : "user_4", "age" : 4 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4651"), "username" : "user_5", "age" : 5 }
{ "_id" : ObjectId("52e2f3b8cdbf70bf8e4652"), "username" : "user_6", "age" : 6 }
```

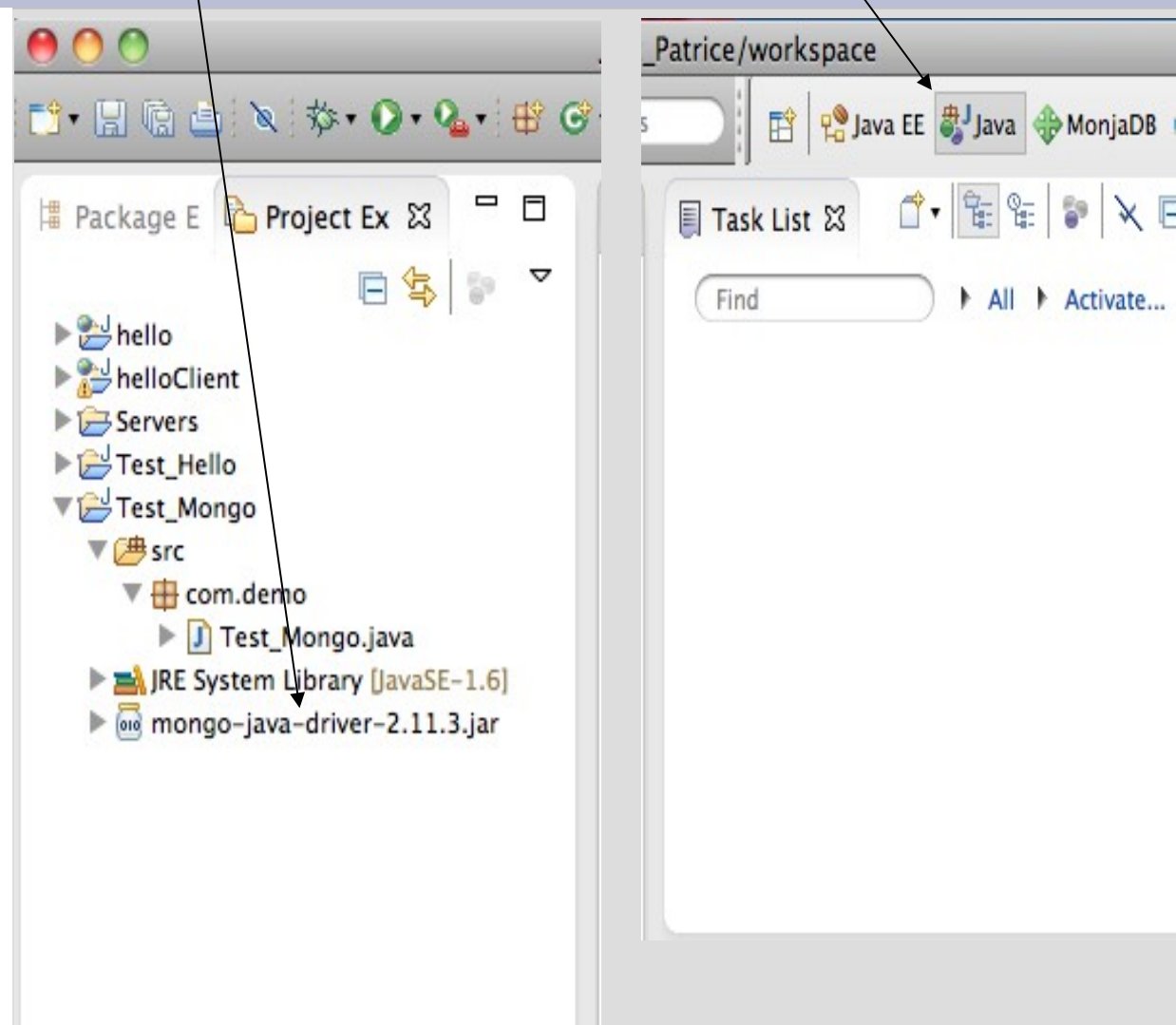
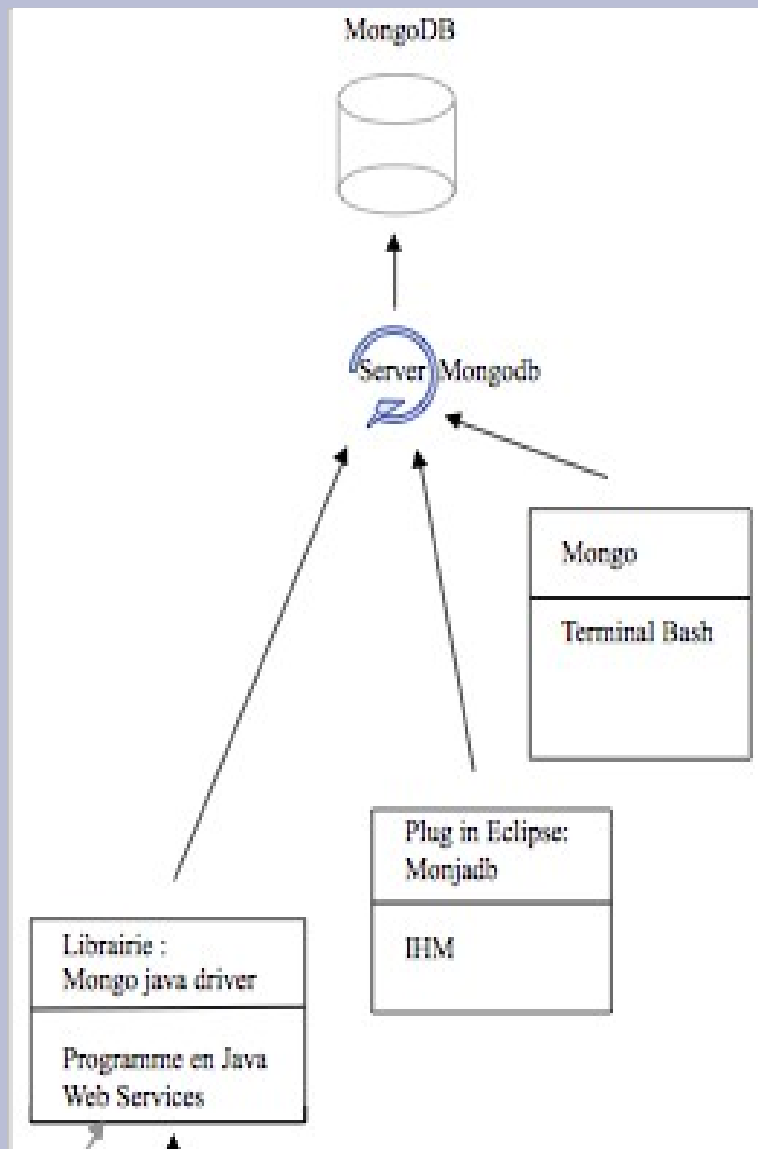
Sommaire



**WebServices et Base MongoDB
avec les outils Eclipse de
développement :
Plugins SoapUI et MonjaDB**



Modification de la base Mongo Avec le driver Java Mongo Perspective Java



<http://hmkcode.com/mongodb-java-simple-example/>

Connexion à la base + Déclaration de variables

```
Hello.java  Web Services Test Client  *Test_Mongo.java  ⌵

package com.demo;

import java.net.UnknownHostException;

/** Java + MongoDB Simple Example */
public class Test_Mongo {

    static String array_names[] = {"John", "Tim", "Brit", "Robin", "Smith", "Lora", "Jer

    static String array_address[][] = {
        {"US", "FL", " Miami"},
        {"US", "FL", " Orlando"},
        {"US", "CA", "San Diego"},
        {"US", "FL", " Orlando"},
        {"US", "FL", " Orlando"},
        {"US", "NY", "New York"},
        {"US", "NY", "Buffalo"},
        {"US", "TX", " Houston"},
        {"US", "CA", "San Diego"},
        {"US", "TX", " Houston"}
    };

    public static void main(String[] args) {
        try {

            // Connect to mongodb
            MongoClient mongo = new MongoClient("localhost", 27017);

            // get database (if database doesn't exists, mongodb will create it for you
            /** Base de données "test" */
            DB db = mongo.getDB("test");

            // get collection
            // if collection doesn't exists, mongodb will create it for you
            /** collection "person" */
            DBCollection collection = db.getCollection("person");
```

Ajout d'enregistrements

```
/***** Ajout d'enreg *****/
// create a document to store key and value
BasicDBObject document ;
String address[];
for(int i = 0 ; i < array_names.length ; i++){
    document = new BasicDBObject();
    //value -> String
    document.append("name", array_names[i]);
    // value -> int
    document.append("age", (int)(Math.random()*80));
    // value -> date
    document.append("join", new Date());

    address = pickAddress();
    // value --> document
    document.append("address", new BasicDBObject("country",address[0])
        .append("state", address[1])
        .append("city", address[2]));

    // value -> array
    document.append("friends", pickFriends());

    collection.insert(document);
}
// get count
System.out.println("Nombre de personnes au total : "+collection.getCount());
// get all document
/**** on affiche toutes les personnes ****/
DBCursor cursor = collection.find();
try {
    while(cursor.hasNext()) {
        System.out.println(cursor.next());
    }
} finally {
    cursor.close();
}
//-----
```

Résultat du programme

Nombre de personnes au total : 10

```
{ "_id" : { "$oid" : "52e58785036474673daf2b48" } , "name" : "Tim" , "age" : 18 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4a" } , "name" : "Robin" , "age" : 10 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4f" } , "name" : "Victor" , "age" : 40 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b47" } , "name" : "John" , "age" : 16 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b49" } , "name" : "Brit" , "age" : 53 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4b" } , "name" : "Smith" , "age" : 48 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4c" } , "name" : "Lora" , "age" : 2 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4d" } , "name" : "Jennifer" , "age" : 32 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4e" } , "name" : "Lyla" , "age" : 27 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b50" } , "name" : "Adam" , "age" : 22 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
```

Nbre de personne de plus de 20 ans = 6

John prend 25 ans...

Nbre de personne de plus de 20 ans = 7

```
{ "_id" : { "$oid" : "52e58785036474673daf2b48" } , "name" : "Tim" , "age" : 18 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4a" } , "name" : "Robin" , "age" : 10 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4f" } , "name" : "Victor" , "age" : 40 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b47" } , "name" : "John" , "age" : 41 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b49" } , "name" : "Brit" , "age" : 53 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4b" } , "name" : "Smith" , "age" : 48 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4c" } , "name" : "Lora" , "age" : 2 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4d" } , "name" : "Jennifer" , "age" : 32 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b4e" } , "name" : "Lyla" , "age" : 27 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
{ "_id" : { "$oid" : "52e58785036474673daf2b50" } , "name" : "Adam" , "age" : 22 , "join" : { "$date" : "2014-01-01T00:00:00Z" } }
```

Done

Query

Update somme

```
// get documents by query
BasicDBObject query = new BasicDBObject("age", new BasicDBObject("$gt",20));

cursor = collection.find(query);
System.out.println("Nbre de personne de plus de 20 ans = "+cursor.count());

/**** Update -- le temps passe pour John 25 ans plus tard****/

// This example show the use of $inc modifier to increase a particular value. Find doc

BasicDBObject newDocument =
    new BasicDBObject().append("$inc", new BasicDBObject().append("age", 25));

//collection.update(new BasicDBObject().append("name", "hostB"), newDocument);
collection.update(new BasicDBObject().append("name", "John"), newDocument);

System.out.println("John prend 25 ans...");

// get documents par un nouveau query
BasicDBObject query2 = new BasicDBObject("age", new BasicDBObject("$gt",20));

cursor = collection.find(query2);
System.out.println("Nbre de personne de plus de 20 ans = "+cursor.count());

//get all again
cursor = collection.find();
try {
    while(cursor.hasNext()) {
        System.out.println(cursor.next());
    }
} finally {
    cursor.close();
}
```

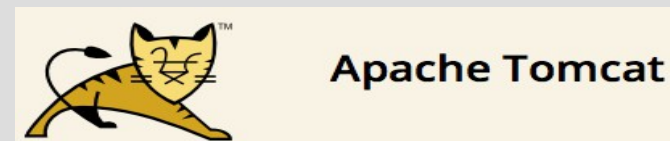
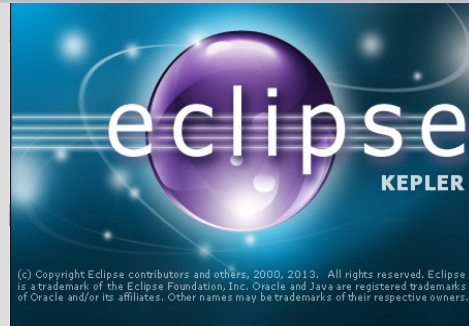
Procédure

```
    /**** Done ****/  
    System.out.println("Done");  
  
    } catch (UnknownHostException e) {  
        e.printStackTrace();  
    } catch (MongoException e) {  
        e.printStackTrace();  
    }  
  
}  
//-----  
//These methods are just used to build random data  
private static String[] pickFriends(){  
    int numberOfFriends = (int) (Math.random()* 10);  
    LinkedList<String> friends = new LinkedList<String>();  
    int random = 0;  
    while(friends.size() < numberOfFriends){  
        random = (int) (Math.random()*10);  
        if(!friends.contains(array_names[random]))  
            friends.add(array_names[random]);  
    }  
    String a[] = {};  
    return friends.toArray(a);  
}  
  
private static String[] pickAddress(){  
    int random = (int) (Math.random()*10);  
    return array_address[random];  
}  
}
```


Résultat procédures

```
"address" : { "country" : "US" , "state" : "FL" , "city" : " Miami"} , "friends" : [ "Lyla" , "John"]}  
, "address" : { "country" : "US" , "state" : "FL" , "city" : " Orlando"} , "friends" : [ "Brit" , "Jennifer" , "Robin"]}  
} , "address" : { "country" : "US" , "state" : "FL" , "city" : " Orlando"} , "friends" : [ "Victor"]}  
, "address" : { "country" : "US" , "state" : "CA" , "city" : "San Diego"} , "friends" : [ "Adam" , "Tim" , "Lora" , "Brit" ,  
, "address" : { "country" : "US" , "state" : "CA" , "city" : "San Diego"} , "friends" : [ "Victor" , "John" , "Robin" , "Lor  
, "address" : { "country" : "US" , "state" : "NY" , "city" : "New York"} , "friends" : [ "Jennifer"]}  
"address" : { "country" : "US" , "state" : "FL" , "city" : " Orlando"} , "friends" : [ "Brit" , "Victor" , "Smith" , "Lora"  
"Z"} , "address" : { "country" : "US" , "state" : "FL" , "city" : " Miami"} , "friends" : [ "Adam"]}  
, "address" : { "country" : "US" , "state" : "FL" , "city" : " Orlando"} , "friends" : [ "John" , "Smith"]}  
, "address" : { "country" : "US" , "state" : "FL" , "city" : " Orlando"} , "friends" : [ "Lyla" , "Victor" , "Jennifer" , "A
```

Conclusion



Eclipse eclipse-jee-kepler-SR1-macosx-cocoa-x86_64
MongoDB 2.4.8
plugin MonjaDB 1.0.2
Library mongo-java-driver-2.11.3
plugin Program SoapUI 4.0.1
Server apache tomcat 6.0.37



